

<i>Title:</i>	Revision of the Bachelor of Science in Chemical Engineering
<i>Sponsor:</i>	<i>Department of Chemical Engineering</i>
<i>Executive Summary:</i>	The Department of Chemical Engineering is proposing to modify its undergraduate program by adding five optional concentrations to the BS degree: 1. Concentration in Polymers and Molecular Engineering 2. Concentration in Energy and Environment 3. Concentration in Nanotechnology 4. Concentration in Process Automation 5. Concentration in Entrepreneurship The BS in chemical engineering major currently has one optional concentration, "Biochemical Engineering". The addition of the five concentrations will strengthen the curriculum by providing more in-depth studies in key areas of chemical engineering; polymers, energy, environment and process automation. The department will also be more competitive with other chemical engineering programs by providing additional topic choices in emerging and multi-disciplinary fields of study; molecular engineering, nanotechnology and entrepreneurship. We anticipate that 20-30% of our students will take one or more of the concentrations.
<i>Description:</i>	Effective Spring 2021, add five concentrations to the Bachelor of Science in Chemical Engineering degree: 1. Concentration in Polymers and Molecular Engineering 2. Concentration in Energy and Environment 3. Concentration in Nanotechnology 4. Concentration in Process Automation 5. Concentration in Entrepreneurship Specific courses will be required to obtain the concentration as shown in the Catalog Statement. Depending on the concentration, the course plan will include; two to four required courses and one to two elective courses chosen from a list specific to each concentration. The concentration course requirements will not increase the total credit hours required for the major for the following concentrations: Process Automation and Entrepreneurship. The concentration course requirements will increase the total credit hours required by 6 for the following concentrations: Polymer & Molecular Engineering, Nanotechnology, Energy & Environment. The concentration title will appear on the student's transcript.
<i>Justification:</i>	The Bachelor of Science in Chemical Engineering has been an official baccalaureate degree in the UIC College of Engineering since 1986. The degree consists of a core set of courses covering; thermodynamics, transport processes (heat transfer, mass transfer, fluid flow), chemical process design, chemical engineering laboratory and professional development. It is an ABET-accredited curriculum preparing students for a career in a wide range of industries, academia and the government. A Biochemical Engineering concentration was added to the Bachelor of Science

degree in 2005.

The core chemical engineering curriculum provides a solid foundation in chemical science and engineering. However, to provide a more enriching experience to our students, there is a need to offer more specialized studies in both traditional and emerging areas of chemical engineering, as well as in engineering-based entrepreneurship. The addition of five new concentrations is being proposed to strengthen the curriculum in key areas of modern chemical engineering research and practice, and in starting an engineering business. The additional courses taken in the different concentrations will provide the students with a greater understanding of chemical engineering principles as well as starting up a company. Ultimately this will lead to more opportunities after graduation, positioning the students to make significant contributions in these multi-disciplinary areas.

The five concentrations being proposed are in areas that the department of chemical engineering has specific expertise and are currently being pursued both in research and in practice; molecular engineering, polymers, nanomaterials, energy, environment, process automation, and entrepreneurship. These specializations play a central role in biological, chemical/industrial, environmental and pharmaceutical processes and technologies and thus are important additions to our curriculum.

Five courses will be added to provide the necessary curriculum covered in the concentrations. CHE 230 Molecular Systems in Chemical Engineering, and CHE 330 Polymer Science will introduce several topics that are important to traditional and emerging fields in chemical engineering; polymers, molecular engineering, macromolecular systems, and nanomaterials. The courses; CHE 454 Molecular and Macromolecular Engineering, and CHE 455 Nanoscale Systems in Chemical Engineering, will expand the list of technical electives in chemical engineering and will provide more comprehensive knowledge of the above topics; polymers, molecular and macromolecular systems, and nanomaterials. CHE 457 Colloidal and Interfacial Phenomena, a technical elective, will also be added to provide comprehensive knowledge in this important field of study.

The five proposed concentrations were discussed at the Chemical Engineering External Advisory Board meeting in June 2019 and were approved by the Board members. The results from a survey of chemical engineering alumni in Summer 2019 also indicated overwhelming support for the five proposed concentrations. In addition, the Concentration in Entrepreneurship was discussed with the Chair in Entrepreneurship of the Department of Managerial Studies in the College of Business Administration, who gave full support of this concentration.

Further, there is a general trend by other chemical engineering programs to include concentrations, specializations, or emphasis programs to the BS degree. For instance, the BS in Chemical Engineering at the University of Illinois Urbana Champaign has two concentrations; chemical engineering and biomolecular engineering; Purdue University has 4 concentrations; biological engineering, energy & environment, materials & polymers, and pharmaceutical engineering; Northwestern University has 6 specializations; bioengineering, chemical process engineering, design, environmental engineering & sustainability, nanotechnology & molecular engineering, and polymer science & engineering; University of Southern California has 6 emphasis programs; biochemical, petroleum, nanotechnology, polymer/materials, environmental, sustainable energy; and the University of Washington Seattle has 7 Focus Areas; Bio & Biomedical Materials, Interfaces & Systems, Data Science, Computation, Statistics & Modeling, Entrepreneurship, Environmental Engineering, Energy Systems, Polymers, and Composites, Colloids &

Interfaces.

Specific information regarding each of the concentrations is given below:

CONCENTRATION IN POLYMERS AND MOLECULAR ENGINEERING (+3 Credits)

Molecular engineering is an interdisciplinary field of study focused on design, characterization, synthesis, assembly, analysis, simulation and applications of molecular-scale systems. Polymer science and engineering is a more traditional area of chemical engineering that is currently not being covered in our curriculum. Molecular engineering includes polymer engineering, interfacial engineering, pharmaceutical engineering, energy harvesting, environmental engineering, nanotechnology, synthetic biology, molecular machines, drug-delivery, self-assembly, and molecular electronics. Molecular engineering is closely related to chemical engineering. Chemical engineers apply chemistry and physics at macro-scale to design processes for large-scale manufacturing of chemicals; while molecular engineers apply chemistry and physics at atomic/molecular scale to design molecules, molecular systems, and molecular assemblies with desired attributes. Since the department of chemical engineering already includes detailed curriculum on chemical processes, the addition of a polymer and molecular engineering concentration will complete the story from the origin of the molecule to its production at large scale with desired properties for their intended applications.

1. REQUIRED COURSES

ChE Selective Courses

- a. CHE 230 Molecular Systems in Chemical Engineering
- b. CHE 330 Polymer Science

ChE Technical Elective

- c. CHE 454 Molecular and Macromolecular Engineering

2. ELECTIVE COURSES

a. Elective Outside the ChE-rubric (Select One)

- i. BIOE 405. Atomic and Molecular Nanotechnology
- ii. PHAR 422. Fundamentals of Drug Action.
- iii. PHAR 423. Biomedical Chemistry
- iv. CHEM 448 Statistical Thermodynamics
- v. PHYS 450/BIOE 450. Molecular Biophysics of the Cell
- vi. CHEM 452/454 Biochemistry
- vii. CHEM 458. Biotechnology and Drug Discovery
- viii. PHYS 461 Thermal and Statistical Physics

b. Second Elective (Select One, +3 Credits):

- i. CHE 425. Nanotechnology for Pharmaceutical Applications
- ii. CHE 438: Computational Molecular Modeling
- iii. CHE 440: Non-Newtonian Fluids
- iv. CHE 453: Electrochemistry
- v. CHE 455 Nanoscale Systems in Chemical Engineering
- vi. CHE 457: Colloidal and Interfacial Phenomena
- vii. 400-level course approved by the Director of Undergraduate Studies
- viii. CHE 392: Undergraduate Research on topic thematically linked with polymer or molecular engineering approved by Chemical Engineering Faculty Member

CONCENTRATION IN NANOTECHNOLOGY (+3 Credits)

1. REQUIRED COURSES

ChE Selective Courses

- a. CHE 230 Molecular Systems in Chemical Engineering
- b. CHE 330 Polymer Science

ChE Technical Elective

- c. CHE 455 Nanoscale Systems in Chemical Engineering

2. ELECTIVE COURSES

a. Elective Outside the ChE-rubric (Select One)

- i. ECE 346. Solid State Device Theory.
- ii. BIOE 405. Atomic and Molecular Nanotechnology
- iii. ME 418. Transport Phenomena in Nanotechnology
- iv. PHYS 431: Modern Physics: Condensed Matter
- v. ECE 440. Nanoelectronics
- vi. ECE 449. Microdevices and Micromachining Technology
- vii. CHEM 458. Biotechnology and Drug Discovery
- viii. BIOE 485. Nanobiosensors

b. Second Elective (Select One, +3 Credits)

- i. CHE 425. Nanotechnology for Pharmaceutical Applications
- ii. CHE 457: Colloidal and Interfacial Phenomena
- iii. 400-level course approved by the director of undergraduate studies
- iv. CHE 392: Undergraduate research on topics thematically linked with nanotechnology, approved by Chemical Engineering Faculty Member

CONCENTRATION IN ENERGY AND ENVIRONMENT (+3 Credits)

1. REQUIRED COURSES

ChE Selective Courses

- a. CHE 230 Molecular Systems in Chemical Engineering
- b. CHE 330 Polymer Science

ChE Technical Elective

- c. CHE 451. Renewable Energy Technologies

Elective Outside the ChE Rubric

- d. CME 322. Environmental Engineering

2. ELECTIVE COURSES (Select One; +3 Credits)

- a. CHE 450: Air Pollution Engineering
- b. 400-level course approved by the director of undergraduate studies
- c. CHE 392: Undergraduate Research on topics thematically linked to sustainability, electrochemistry, water science, or renewable resources, approved by Chemical Engineering Faculty Member

CONCENTRATION IN PROCESS AUTOMATION (+0 Credits)

1. REQUIRED COURSES

ChE Technical Elective

- a. CHE 433. Process Simulation with Aspen Plus (Counted as technical elective)

2. ELECTIVE COURSES

Elective from outside ChE-rubric (Select One)

- a. ECON 120 and ECON 220 Theory and Applications of Microeconomics

	b. IE 201 Financial Engineering c. MATH 310 Applied Linear Algebra d. MATH 419 Models in Applied Mathematics <u>CONCENTRATION IN ENTREPRENEURSHIP (+ 6 Credits)</u> 1. REQUIRED COURSES (+ 3 Credits) <u>ChE Technical Elective</u> a. 400-level course approved by the director of undergraduate studies <u>Two Electives outside the ChE Rubric (+3 Credits)</u> a. ENTR 310 Introduction to Entrepreneurship b. ENTR 200 Survey of Entrepreneurship 2. ELECTIVE COURSES (Select one, + 3 Credits) i. MENG 401 Engineering Management ii. MENG 402 Intellectual Property Law iii. MENG 404 Entrepreneurship. iv. ENTR 445 New Venture Planning
<i>Catalog Statement:</i>	<i>See below</i>
<i>Minority Impact Statement:</i>	<i>The changes will clarify and improve the degree requirements for all students and will not adversely impact minority students.</i>
<i>Budgetary and Staff Implications:</i>	<i>None expected</i>
<i>Library Resource Implications:</i>	<i>None expected</i>
<i>Space Implications:</i>	<i>None expected</i>
<i>Unit (e.g. department) approval date:</i> <i>College (educational policy committee, faculty) approval dates:</i>	Provide unit dates, including department, college/school committees, faculty Approvals, as applicable. Chemical engineering faculty and department approval: 05/13/19 faculty meeting Education Policy Committee approval:

<i>Contact Person:</i>	Alan Zdunek zdunek@uic.edu
<i>Proposed Effective Date/Term:</i>	Fall 2020

Current	Revised
BS in Chemical Engineering	BS in Chemical Engineering
<i>Degree Requirements</i> To earn a Bachelor of Science in Chemical Engineering degree from UIC, students need to complete University, college, and department degree requirements. The Department of Chemical Engineering degree requirements are outlined below. Students should consult the <i>College of Engineering</i> section for additional degree requirements and college academic policies.	SAME
BS in Chemical Engineering Degree Requirements Hours	SAME
Nonengineering and General Education Requirements 70	
Required in the College of Engineering 49	
Technical Elective ^a 3	
Selective 3	
Electives outside the Major Rubric ^a 3	
Total Hours—BS in Chemical Engineering 128	
 <i>Nonengineering and General Education Requirements</i>	SAME
ENGL 160— Academic Writing I: Writing for Academic and Public Contexts 3	SAME
ENGL 161— Academic Writing II: Writing for Inquiry and Research 3	
Exploring World Cultures course ^a 3	
Understanding the Creative Arts course ^a 3	
Understanding the Past course ^a 3	SAME
Understanding the Individual and Society course ^a 3	
Understanding U.S. Society course ^a 3	
MATH 180—Calculus I ^b 4	
MATH 181—Calculus II ^b 4	
MATH 210—Calculus III ^b 3	
MATH 220—Introduction to Differential Equations 3	
PHYS 141—General Physics I (Mechanics) ^b 4	
PHYS 142—General Physics II (Electricity and Magnetism) ^b 4	
CHEM 122- General College Chemistry I ^b (Lecture) 4	
CHEM 123- General College Chemistry I ^b (Laboratory) 1	
OR CHEM 116 – Honors and Major General and Analytical Chemistry I 5	
CHEM 124—General College Chemistry II ^b (Lecture) 4	
CHEM 125—General College Chemistry II ^b (Lab) 1	
OR CHEM 118 – Honors and Major General and Analytical Chemistry II 5	
	SAME

CHEM 222—Analytical Chemistry ^c	4	
CHEM 232—Organic Chemistry I	4	
CHEM 233- Organic Chemistry Laboratory	2	
CHEM 234—Organic Chemistry II	4	
CHEM 342—Physical Chemistry I	3	
Total Hours- Nonengineering and General Education Requirements	70	
a Students should consult the General Education section of the catalog for a list of approved courses in this category.		
b This course is approved for the Analyzing the Natural World General Education category.		
^c Students who take CHEM 116 and CHEM 118 to fulfill the general chemistry requirement do not need to take CHEM 222. Instead they should enroll in one of the following: CHEM 314: Inorganic Chemistry; CHEM 452: Biochemistry; or CHEM 402: Chemical Information Systems and CHEM 444: Physical Chemistry III.		
Selective: Choose from CHE 330 and CHEM 346 (3)		
Total Hours: Selective	3	
Required in the College of Engineering		
Courses	Hours	
ENGR 100—Orientation ^a	0 ^a	
CHE 201—Introduction to Thermodynamics	3	
CHE 205—Computational Methods in Chem. Eng.	3	
CHE 210—Material and Energy Balances	4	
CHE 301—Chemical Engineering: Thermodynamics	3	
CHE 311—Transport Phenomena I	3	
CHE 312—Transport Phenomena II	3	
CHE 313—Transport Phenomena III	3	
CHE 321—Chemical Reaction Engineering	3	SAME
CHE 341—Chemical Process Control	3	
CHE 381—Chemical Engineering Laboratory I	2	
CHE 382—Chemical Engineering Laboratory II	2	
CHE 396—Senior Design I	4	
CHE 397—Senior Design II	4	SAME
CHE 499—Professional Development Seminar	0	
ECE 210—Electrical Circuit Analysis	3	
CME 260—Properties of Materials	3	
CS 109—C/C++ Programming for Engineers with MatLab	3	
Total Hours—Required in the College of Engineering	49	
a ENGR 100 is one-semester-hour course, but the hour does not count toward the total hours required for graduation.		

BS in Chemical Engineering, Bioengineering Concentration

Students in this concentration complete the following:

Required Courses ^a

Technical Elective

CHE 422	Biochemical Engineering	3
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Electives

Select two electives in nonmajor rubric category from the following:	5-7
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BIOS 350	General Microbiology
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BIOS 351	Microbiology Laboratory
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CHEM 352	Introductory Biochemistry
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CHEM 452	Biochemistry I
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Total Hours	8-10
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^a Due to structure of the concentration and the prerequisites required for some of the courses, students in the concentration will be required to take a minimum of 130 semester hours for the degree.

SAME

BS in Chemical Engineering, Energy & Environment Concentration

Students in this concentration complete the following:

Code	Title	Hours
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Required Courses ^a

CHE 230	Molecular Systems in Chemical Engineering	3
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CHE 330	Polymer Science	3
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Technical Electives

CHE 451	Renewable Energy Technologies	3
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Electives

Electives outside the Major Rubric:

CME 322	Environmental Engineering	3
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Second Technical Elective (Select one)	3
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A 400-level course approved by the director of undergraduate studies

CHE 392	Undergraduate Research Topics Thematically linked to Sustainability, Electrochemistry, Water Science and Renewable Resources (approved by Chemical Engineering Faculty Member)
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CHE 450	Air Pollution Engineering
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CHE 453	Electrochemistry
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Total Hours	15
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^a Due to structure of the concentration and the prerequisites required for some of the courses, students in the concentration will be required to take a minimum of 134 semester hours for the degree (+6 credit hours)

BS in Chemical Engineering, Entrepreneurship Concentration

Students in this concentration complete the following:

Code	Title	Hours
Required Courses ^a		
Technical Elective		
	A 400-level course approved by the director of undergraduate studies	3
Electives		
Electives outside the Major Rubric:		
ENTR 200	Survey of Entrepreneurship	3
ENTR 310	Introduction to Entrepreneurship	3
Second Electives outside the Major Rubric (Select one)		3
ENGR 401	Engineering Management	
ENGR 402	Intellectual Property Law	
ENGR 404	Entrepreneurship	
ENTR 445	New Venture Planning	
Total Hours		12

a Due to structure of the concentration and the prerequisites required for some of the courses, students in the concentration will be required to take a minimum of 134 semester hours for the degree (+6 credit hours)

BS in Chemical Engineering, Nanotechnology Concentration

Students in this concentration complete the following:

Code	Title	Hours
Required Courses ^a		
CHE 230	Molecular Systems in Chemical Engineering	3
CHE 330	Polymer Science	3
Technical Electives		
CHE 455	Nanoscale Systems in Chemical Engineering	3
Electives		
Electives outside the Major Rubric:		3-4
ECE 346	Solid State Device Theory	
BIOE 405	Atomic and Molecular Nanotechnology	
ME 418	Transport Phenomena in Nanotechnology	
ECE 440	Nanoelectronics	

	ECE 449	Microdevices and Micromachining Technology	
	CHEM 458	Biotechnology and Drug Discovery	
	BIOE 485	Biosensors	
	Second Technical Elective (Select one)		3
	A 400-level course approved by the director of undergraduate studies		
	CHE 392	Undergraduate Research Topics (Thematically linked to Nanotechnology)	
	CHE 425	Nanotechnology for Pharmaceutical Applications	
	CHE 457	Colloidal and Interfacial Phenomena	
	Total Hours		15-16
	^a Due to structure of the concentration and the prerequisites required for some of the courses, students in the concentration will be required to take a minimum of 134 semester hours for the degree (+6 credit hours)		
	BS in Chemical Engineering, Polymer & Molecular Engineering Concentration Students in this concentration complete the following:		
	Code	Title	Hours
	Required Courses ^a		
	CHE 230	Molecular Systems in Chemical Engineering	3
	CHE 330	Polymer Science	3
	Technical Electives		
	CHE 454	Molecular and Macromolecular Engineering	3
	Electives		
	Electives outside the Major Rubric:		3-4
	BIOE 405	Atomic and Molecular Nanotechnology	
	PHAR 422	Fundamentals of Drug Action	
	PHAR 423	Biomedical Chemistry	
	CHEM 448	Statistical Thermodynamics	
	BIOE 450	Molecular Biophysics of the Cell	
	CHEM 452	Biochemistry	
	BIOE 485	Biosensors	
	Second Technical Elective (Select one)		3
	A 400-level course approved by the director of undergraduate studies		

	<table><tr><td>CHE 392</td><td>Undergraduate Research Topics (Thematically linked to Polymer Science or Molecular Engineering)</td><td></td></tr><tr><td>CHE 450</td><td>Air Pollution Engineering</td><td></td></tr><tr><td>CHE 453</td><td>Electrochemistry</td><td></td></tr><tr><td>Total Hours</td><td></td><td>15-16</td></tr></table>	CHE 392	Undergraduate Research Topics (Thematically linked to Polymer Science or Molecular Engineering)		CHE 450	Air Pollution Engineering		CHE 453	Electrochemistry		Total Hours		15-16																								
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CHE 453	Electrochemistry																																				
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	BS in Chemical Engineering, Process Automation Concentration Students in this concentration complete the following: <table><tr><th>Code</th><th>Title</th><th>Hours</th></tr><tr><td colspan="3">Required Courses ^a</td></tr><tr><td colspan="3">Technical Elective</td></tr><tr><td>CHE 433</td><td>Process Simulation with Aspen Plus</td><td>3</td></tr><tr><td colspan="3">Electives</td></tr><tr><td colspan="2">Electives outside the Major Rubric:</td><td>3</td></tr><tr><td>ECON 120</td><td>Principles of Microeconomics</td><td></td></tr><tr><td>IE 201</td><td>Financial Engineering</td><td></td></tr><tr><td>ECON 220</td><td>Microeconomics: Theory and Applications</td><td></td></tr><tr><td>MATH 310</td><td>Applied Linear Algebra</td><td></td></tr><tr><td>MATH 419</td><td>Models in Applied Mathematics</td><td></td></tr><tr><td>Total Hours</td><td></td><td>6</td></tr></table>	Code	Title	Hours	Required Courses ^a			Technical Elective			CHE 433	Process Simulation with Aspen Plus	3	Electives			Electives outside the Major Rubric:		3	ECON 120	Principles of Microeconomics		IE 201	Financial Engineering		ECON 220	Microeconomics: Theory and Applications		MATH 310	Applied Linear Algebra		MATH 419	Models in Applied Mathematics		Total Hours		6
Code	Title	Hours																																			
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MATH 310	Applied Linear Algebra																																				
MATH 419	Models in Applied Mathematics																																				
Total Hours		6																																			
	^a Due to structure of the concentration and the prerequisites required for some of the courses, students in the concentration will be required to take a minimum of 128 semester hours for the degree (+0 credit hours)																																				

Sample Course Schedule		
Freshman Year		
First Semester	Hours	
MATH 180—Calculus I	4	
CHEM 122—General College Chemistry I (Lecture)	4	
CHEM 123—General College Chemistry I (Lab)	4	1
<u>OR</u> CHEM 116 – Honors and Major General and Analytical Chemistry I	5	
ENGL 160—Academic Writing I: Writing for Academic and Public Contexts	3	
CS 109—C/C++ Programming for Engineers with MatLab	3	
ENGR 100—Orientation ^a	1 ^a	
Total Hours	15	
Second Semester	Hours	
MATH 181—Calculus II	4	
PHYS 141—General Physics I (Mechanics)	4	
ENGL 161—Academic Writing II: Writing for Inquiry and Research	3	
CHEM 124—General College Chemistry II (Lecture)	4	
CHEM 125—General College Chemistry II (Lab)	1	
<u>OR</u> CHEM 118– Honors and Major General and Analytical Chemistry II	5	
Total Hours	16	
		SAME
Sophomore Year		
First Semester	Hours	
MATH 210—Calculus III	3	
PHYS 142—General Physics II (Electricity and Magnetism)	4	
CHEM 232—Organic Chemistry I	4	
CHE 201—Introduction to Thermodynamics	3	
General Education Core course	3	
Total Hours	17	
		SAME
Second Semester	Hours	
MATH 220—Introduction to Differential Equations	3	
CHEM 234—Organic Chemistry II	4	
CHEM 233—Organic Chemistry Lab I	2	
CHE 205—Computational Methods in Chem. Eng.	3	
CHE 210—Material and Energy Balances	4	
CME 260— Properties of Materials	3	
		SAME
Total Hours	19	
Junior Year		
First Semester	Hours	
ECE 210—Electrical Circuit Analysis	3	

CHE 301—ChE Thermodynamics	3	
CHE 311—Transport Phenomena I	3	
CHEM 222—Analytical Chemistry	4	
General Education Core course	3	
Total Hours	16	
Second Semester	Hours	
CHEM 342—Physical Chemistry I	3	
CHE 312—Transport Phenomena II	3	
CHE 313—Transport Phenomena III	3	
CHE 321—Chemical Reaction Engineering	3	
General Education Core course	3	
Total Hours	15	
Senior Year		
First Semester	Hours	
CHE 381—Chemical Engineering Laboratory I	2	
CHE 396—Senior Design I	4	
CHE 330 or CHEM 346	3	
CHE Technical elective—Selected from CHE 410, 413, 421, 422, 423, 431, 438, 440, 441, 445, 450, 456, 494, or 392 (departmental approval is required for CHE 392)	3	
General Education Core Course	3	
Total Hours	15	
Second Semester	Hours	
CHE 382—Chemical Engineering Laboratory II	2	
CHE 341—Chemical Process Control	3	
CHE 397—Senior Design II	4	
CHE 499—Professional Development Seminar	0	
Elective outside the major rubric	3	
General Education Core course	3	
Total Hours	15	
		Same